Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 1 of 9 · find part c in this mark scheme extract

4(a)(i)	A1	Award **one** mark for: • (At the highest level) in the main scope/program/file (1)	Java and C#: allow answers that indicate inside namespace e.g. 'before main class'	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 2 of 9 · find part c in this mark scheme extract

4(a)(ii)	A2	Award **one** mark for: • Only from within the scope/subprogram/code block in which it is created/declared (1)		1

Question	mp	Answer	Additional guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 3 of 9 · find part c in this mark scheme extract

4(b)	B1 B2	Award **two** marks for a linked explanation such as: • **Figure 4** executes fewer comparisons/tests (1), because (the use of nested ifs/else ifs/ means that) as soon as a test evaluates to true, all the remaining tests are skipped (1) • **Figure 5** executes more comparison/tests (1), because every if statement must be executed, even if any test before has already evaluated to true (1)		**2**

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 4 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 5 of 9 · find part c in this mark scheme extract

4(c)		Award **one** mark for each:		
	C1	Subprogram must attempt to use both pString and pNum to generate Key (1)	Award attempted use, even if not correct	
	C2	1D indexing used to access chars in string (1)	Can be anywhere in code (main or subprogram) Any method e.g. • pString[0]	
	C3	2 pairs of separated chars obtained (1)	First 2 **and** last 2 characters obtained from pString Any method Can be anywhere in code • pString[0] + pString[1] • pString[2] + pString[3] • pString[0:2] • pString[2:4]	
	C4	Key concatenated correctly (1)	Key must be correct Can be anywhere in code	
	C5	Subprogram must return their key to the calling line (1)	Do not award return without a value • Value must be variable/value intended to be key	
	C6	Complete line to call new subprogram (1)	Must be fully correct `myNewKey = genNewKey(myString, myNumber)`	**6**

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 6 of 9 · find part c in this mark scheme extract

C#

```
// Q04cFINISHED

namespace Q04cFINISHED
{
    class Q04cFINISHED
    {
        static String genNewKey (String pString, int pNum)
        {
            String newKey;

            // Write your code below this line
            newKey = pString.Substring (0, 2);            // First two chars
            newKey = newKey + pNum.ToString();            // Number
            newKey = newKey + pString.Substring (2, 2);   // Last two chars

            return (newKey);
        }

        static void Main(string[] args)
        {
            String myNewKey = "";
            int myNumber = 0;
            String myString = "";

            Console.WriteLine ("Enter a string:");
            myString = Console.ReadLine();
            Console.WriteLine ("Enter a whole number: ");
            myNumber = Convert.ToInt32(Console.ReadLine());

            if (myString.Length != 4)
            {
                Console.WriteLine ("String must be four characters");
            }
            else
            {
                Console.WriteLine ("Original: " + myString +
                        " " + myNumber.ToString());

                // Complete the call to the subprogram
                myNewKey = genNewKey (myString, myNumber);
                Console.WriteLine ("New: " + myNewKey);
            }
        }
    }
}
```

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 7 of 9 · find part c in this mark scheme extract

Java

```java
// Q04cFINISHED

import java.util.Scanner;

public class Q04cFINISHED
{
    static String genNewKey (String pString, int pNum)
    {
        String newKey;

        // Write your code below this line
        newKey = pString.substring(0,2);  // First two chars
        newKey = newKey + String.valueOf (pNum);              // Number
        newKey = newKey + pString.substring (2, 4);   // Last two chars

        return (newKey);
    }

    public static void main(String[] args)
    {
        String myNewKey = "";
        int myNumber = 0;
        String myString = "";
        Scanner myKeyboard = new Scanner (System.in);

        System.out.println ("Enter a string:");
        myString = myKeyboard.nextLine();
        System.out.println ("Enter a whole number: ");
        myNumber = myKeyboard.nextInt();

        if (myString.length() != 4)
        {
            System.out.println ("String must be four characters");
        }
        else
        {
            System.out.println ("Original: " + myString +
                                " " + String.valueOf(myNumber));

            // Complete the call to the subprogram
            myNewKey = genNewKey (myString, myNumber);
            System.out.println ("New: " + myNewKey);
        }
    }
}

```

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 8 of 9 · find part c in this mark scheme extract

Python

```python
# Q04cFINISHED

def genNewKey (pString, pNum):
    newKey = ""         # Make new word here

    # Write your code below this line
    newKey = pString[0:2]               # First two chars
    newKey = newKey + str (pNum)        # Number
    newKey = newKey + pString[2:4]      # Last two chars
    return (newKey)

myString = input ("Enter a string: ")
myNumber = int (input ("Enter a whole number: "))

if (len (myString) != 4):
    print ("String must be four characters")
else:
    print ("Original: ", myString, myNumber)

    # Complete the call to the subprogram
    myNewKey = genNewKey (myString, myNumber)
    print ("New word:", myNewKey)

```

Type MS · Subprograms and strings: complete the function and its call

Answer 1 of 6 · Question 4 part c · 6 marks

2024 JUN · Paper 02 · extract 9 of 9 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 2 of 6 · Question 4 part b · 7 marks

2020 NOV · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

4 (a)	A1	2.5.2	english < 40, maths < 50 (1)		
	A2	2.5.3	Use of **and** operator in IF (1)		
	A3	2.5.2	english < 40, maths < 50 (1)		
	A4	2.5.3	Use of **or** operator in 'else if 1' (1)		
	A5	2.5.2	english >= 80 AND maths >=85 (1)		
	A6	2.5.3	Correct completion of final else with messageIndex = 3 (1)		(6)

Code example

Python

```
if (english < 40 and maths < 50):
    messageIndex = 0
elif (english < 40 or maths < 50):
    messageIndex = 1
elif (english >= 80 and maths >= 85):
    messageIndex = 2
else:
    messageIndex = 3
```

Type MS · Subprograms and strings: complete the function and its call

Answer 2 of 6 · Question 4 part b · 7 marks

2020 NOV · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 2 of 6 · Question 4 part b · 7 marks

2020 NOV · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

4 (b)	B1	2.2.2	Loop checks each pupil for low attendance		(7)
	B2	2.1.6	Display of name of low attendances		
	B3	2.3.1	Counter for high attenders initialised		
	B4	2.2.2	Loop uses correct comparison (>=)		
	B5	2.6.2	Number of high attenders displayed correctly		
	B6	2.6.3	Subprogram for option 1 called correctly		
	B7	2.6.3	Subprogram for option 2 called correctly		

Code example

Python

```
#   write subprogram for menu option 1 here
def lowAttendance():
    for x in range(len(pupilAttendance)):
        if pupilAttendance[x-1][2] < 75 :
            print (pupilAttendance[x-1][0], pupilAttendance[x-1][1])
    return

#   write subprogram for menu option 2 here

def highAttendance():
    count = 0
    for x in range(len(pupilAttendance)):
        if pupilAttendance[x-1][2] >= 90 :
            count += 1
    return count
```

```
Faroukh Salah
Amara Grzinski
Taz Grimstow
Sadia Bhatti
Fernado Askabat
Siyao Wang
```

```
if option == 1:
    # complete if statement
    lowAttendance()

elif option == 2:
    # complete else if statement
    highAtt = highAttendance()
    print ("There were ", highAtt," high attenders")
```

```
There were  7  high attenders
```

13 of 16

Type MS · Subprograms and strings: complete the function and its call

Answer 2 of 6 · Question 4 part b · 7 marks

2020 NOV · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 1 of 7 · find part a in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Mark
4 (b)(i)	B1	Award up to 2 marks for a linked explanation: • binary search can be quicker / is more effective than a linear search / is more effective with larger lists (1) as it does not have to examine each item in the list (1) • binary search halves the list each time (1) so it can be faster to find an item (1) • binary search requires fewer comparisons than a linear search to establish an item is not in the list (1) because the linear list		

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 2 of 7 · find part a in this mark scheme extract

would need to compare each item before establishing this (1) (2)

4 (b)(ii) Correct answer

Position in list	Product code	Order examined
1	ark11	
2	asp11	
3	bar13	
4	dri15	1
5	mil19	
6	rib10	2
7	str15	3
8	tor16	

Award one mark for each correct value in order column (4)

B1	Start of search correct	Accept 5 and 7 for B1 and B2 (2 marks)
B2	Second search item correct	Follow through if start of search incorrect
B3	Third search item correct	Follow through if start of search incorrect
B4	All correct	

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 3 of 7 · find part a in this mark scheme extract

		All correct		
4 (b)(iii)	B5	Award 1 mark for: 3 or log n + 1		(1)

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 4 of 7 · find part a in this mark scheme extract

		or $\log_2 n + 1$		(1)
4 (b)(iv)	B6	Award 1 mark for any of: • bubble sort • merge sort	Accept any known sorting algorithm	(1)

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 5 of 7 · find part a in this mark scheme extract

Python Code Mark Scheme

Question		Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 6 of 7 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	A1	At least 1 variable has a meaningful name	Ignore spelling mistakes in input message	(7)
	A2	Product name requested using a suitable input message		
	A3	Random number generated that would be at least 10 **or** no higher than 30		
	A4	Random number generated that would be in the correct range 10 to 30		
	A5	First 3 letters of product name generated		
	A6	First 3 letters of product name and random number concatenated to generate productCode		
	A7	productCode and productName output in the same print statement		
Code examples				
Python				

```
# Get input
productName = input("Enter the product name: ")

randomNum = 0

# Generate a random number between 10 and 30 inclusive
randomNum = random.randint(10,30)

# Generate the product code - first three letters of p
productCode = productName[0:3] + str(randomNum)

# Display the product code and the product name
print(productCode + " " + productName)
```

Type MS · Subprograms and strings: complete the function and its call

Answer 3 of 6 · Question 4 part a · 7 marks

2021 JUN · Paper 02 · extract 7 of 7 · find part a in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	...	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 1 of 13 · find part a in this mark scheme extract

4 (b)	B1 B2	**Award up to 2 marks for a linked description such as:** • To check to see if a word is the same when it is reversed / is a palindrome (1) and output an appropriate message / repaper is the same when reversed (1)		(2)

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 2 of 13 · find part a in this mark scheme extract

Coding

Question	mp	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 3 of 13 · find part a in this mark scheme extract

4 (a)	**Award one mark for each of:**			
		Validation		
	A1	Suitable input prompt and storing it in binaryPattern (1)		
	A2	Length of binary pattern validated as 8 (1)		
	A3	Loop used to check the length of binary pattern (1)		
		Binary to denary		
	A4	Loop through each digit in the binary pattern (1)		
	A5	Correct conversion of at least one binary digit to relevant placeholder (1)		
	A6	Attempt at running total using denaryNumber (1)		
	A7	Correct running total using denaryNumber (1)		
	A8	Print statement includes the binary pattern and the denary number (1)		
	A9	Print statement outside of conversion loop (1)		
	A10	Comment explains how conversion works (1)		
	A11	Program executes correctly for any binary pattern that is exactly 8 characters long only (1)		(11)
Code examples				

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 4 of 13 · find part a in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 5 of 13 · find part a in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 6 of 13 · find part a in this mark scheme extract

For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

Question	mp	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 7 of 13 · find part a in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 8 of 13 · find part a in this mark scheme extract

```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 9 of 13 · find part a in this mark scheme extract

Java

```
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 10 of 13 · find part a in this mark scheme extract

```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 11 of 13 · find part a in this mark scheme extract

Python

```python
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 12 of 13 · find part a in this mark scheme extract

```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("-------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # -----------------------------------------------------------------------------------
72
73      # End the program when there are no more words
74      print("End of program")
```

Type MS · Subprograms and strings: complete the function and its call

Answer 4 of 6 · Question 4 part a · 11 marks

2021 NOV · Paper 02 · extract 13 of 13 · find part a in this mark scheme extract

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Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part b in this mark scheme extract

4(a)(i)	A1 A2	Award **one** mark for each correct cell: **Plaintext** / **Shift** / **Ciphertext** PIXEL / -4 / LETAH CLOUD / +3 / FORXG	Allow lower case ciphertext	**2**

Question	mp	Answer	Additional Guidance	Mark
4(a)(ii)	A3 A4	Award **two** marks for a comparison such as: • Both produce same result (IQNF)(1) because the shift of -6 followed by +8 is the same as the shift of +2 (1) • The result is the same (IQNF) (1), but the single shift is more efficient than a double shift (1) Practical explanation: GOLD (-6) = AIFX (+8) = IQNF (1) GOLD (+2) = IQNF (1)	can have 1 mark from MP and 1 mark from Practical e.g. GOLD (+2) = IQNF (1) which would be same result (1)	**2**
4(a)(iii)	A5 A6	Award **two** marks for a linked explanation such as: The letter Y has been encrypted incorrectly as the letter V / the letter Y should have been encrypted to the letter D (1), because the +5 shift did not roll over the end of the alphabet correctly (1)	identify where error occurred (1) and can explain how error occurred (1)	**2**

Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
4(b)	Award **one** mark for each of:		B1 does not require casting for data types on input B6 do **not** accept taking just first digit from a string B8 is only for print(), key may not be correct	
	B1	Word and 2 numbers input		
	B2	Selection/loop used to check word input		
	B3	Test for word length exactly 2		
	B4	Appropriate error message for invalid string input		
	B5	Reverse the inputted word, any method		
	B6	Whole number part of the decimal number generated		
	B7	Correct key generated (concatenation)		
	B8	Key output		**8**

Suggested test data and output			
word	**int**	**float**	**Output**
qwe/q			error message
qw	12	17.89	12wq17

Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part b in this mark scheme extract

C#

```
// Create the variables
String myWord = "";
String newWord = "";
int myNum = 0;
double myDecimal = 0.0;
String myKey = "";
int myWhole = 0;

// Take a word as input
Console.Write("Enter a word: ");
myWord = Console.ReadLine();

// Check that the word is correct length
if (myWord.Length == 2)
{
    // Take a whole number as input
    Console.Write("Enter a whole number: ");
    myNum= Convert.ToInt32(Console.ReadLine());

    // Take a decimal number as input
    Console.Write("Enter a decimal number: ");
    myDecimal = Convert.ToDouble(Console.ReadLine());

    // Reverse the letters in the word
    newWord = myWord[1].ToString() + myWord[0].ToString();

    // Find the whole number part of the decimal number
    myWhole = (int)myDecimal;

    // Create the new key
    myKey = String.Concat(myNum, newWord, myWhole);

    // Display the new key for the user
    Console.Write("The key is " + myKey);
}
else
{
    // Word is not the correct length, so display an error message
    Console.WriteLine("Invalid word entered");
}
```

Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part b in this mark scheme extract

Java

```
    // Write your code below this line

    Scanner myScanner = new Scanner (System.in);

    // Create the variables
    String myWord = "";
    String newWord = "";
    int myNum = 0;
    Double myDecimal = 0.0;
    String myKey = "";
    int myWhole = 0;

    // Take a word as input
    System.out.print("Enter a word: ");
    myWord = myScanner.nextLine();

    // Check that the word is correct length
    if (myWord.length() == 2)
    {
        // Take a whole number as input
        System.out.print("Enter a whole number: ");
        myNum = Integer.parseInt(myScanner.nextLine());

        // Take a decimal number as input
        System.out.print("Enter a decimal number: ");
        myDecimal = Double.parseDouble (myScanner.nextLine());

        // Reverse the letters in the word
        newWord = Character.toString(myWord.charAt(1)) +
                  Character.toString(myWord.charAt(0));

        // Find the whole number part of the decimal number
        myWhole = myDecimal.intValue();

        // Create the new key
        myKey = myNum + newWord + myWhole;

        // Display the new key for the user
        System.out.println("The key is " + myKey);
    }
    else
    {
        // Word is not the correct length, so display an error message
        System.out.println("Invalid word entered");
    }
}
```

Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part b in this mark scheme extract

Python

```
# Take a two-letter word as input
myWord = input ("Enter a word: ")

# Check that the word is correct length
if (len (myWord) == 2):

    # Take a whole number as input
    myNum = int (input ("Enter a whole number: "))

    # Take a decimal number as input
    myDecimal = float (input ("Enter a decimal number: "))

    # Reverse the letters in the word
    newWord = myWord[1] + myWord[0]

    # Find the whole number part of the decimal number
    myWhole = int (myDecimal)

    # Create the new key
    myKey = str (myNum) + newWord + str (myWhole)

    # Display the new key for the user
    print (myKey)

# Word is not the correct length, so display an error message
else:
    print ("Invalid word entered")
```

Type MS · Subprograms and strings: complete the function and its call

Answer 5 of 6 · Question 4 part b · 8 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part b in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Subprograms and strings: complete the function and its call

Answer 6 of 6 · Question 2 part c.ii · 4 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c.ii in this mark scheme extract

2(a)	A1	**The only correct answer is D** ***A*** *is not correct because abstraction is a technique used in computational thinking* ***B*** *is not correct because encryption is a security measure* ***C*** *is not correct because iteration is a programming construct*		**1**

Question	mp	Answer	Additional guidance	Mark
2(b)(i)		Award **one** mark for each correct cell:		
	B1 B2	real/double/float: (1) • pay rate/18.75 (1) • total pay/356.25 (1) • tax/0.2/71.25 (1) • income/285.00 (1) int/integer: (1) • hours/19 (1)	Example must match data type for B2 Award suitable example even if not taken from payslip Accept string and matching example e.g. Hours	**2**

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Type MS · Subprograms and strings: complete the function and its call

Answer 6 of 6 · Question 2 part c.ii · 4 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
2(b)(ii)		Award **one** mark for each correct cell:		
	B3	**Input:** • Pay Rate/18.75 (1) • Hours/19 (1)		
	B4	**Process**: • Pay Rate * Hours/18.75 * 19 (1) • Total Pay * Tax/356.25 * 0.2 (1) • Total Pay – Tax/356.25 – 71.25 (1)	Don't need output e.g. total pay, just the calculation part e.g. pay rate * hours: Total pay = pay rate * hours Accept 0.2 or 20% as part of calculation for tax Ok to mix types e.g. Total Pay * 20% Accept descriptions such as *multiply the pay rate by hours (hours etc could be taken from input box/ B3 mark)*	
	B5	**Output:** • Total Pay/356.25 (1) • Tax/71.25 (1) • Income/285 (1)		**3**

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Type MS · Subprograms and strings: complete the function and its call

Answer 6 of 6 · Question 2 part c.ii · 4 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
2(c)(i)	C1 C2	Award **one** mark for **one** correct cell: Award **two** marks for all **three** correct cells: • **Normal:** any value between 0-255 e.g. 125 • **Boundary:** 0/255 • **Erroneous:** any negative value/any value 256 or above	Accept -1/256 for boundary data Don't accept non-int types	**2**

Question	mp	Answer	Additional Guidance	Mark
2(c)(ii)		Award **one** mark for any of the following up to a maximum of **four** marks:		
	C3	Input changed to int before comparison (1)		**4**
	C4	Relational operator used correctly to check for valid input (both **>=**0 and **<**256 or equivalent) (1)	Allow: >0 and/or <255	
	C5	'and' logical operator used to combine relational tests (1)	or operator acceptable if candidate rearranges the selection statement and test works for correct code option – check if C4 is now awardable	
	C6	Calling of `getHex(userInput)` (1)	Call must be inside print ( ) Subprogram and argument must be functionally correct Ignore additional variables/strings in print statement	

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Type MS · Subprograms and strings: complete the function and its call

Answer 6 of 6 · Question 2 part c.ii · 4 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c.ii in this mark scheme extract


```
# complete the line to convert the user input to integer
userInput = int(input("please enter a number between 0 and 255 \n"))

# complete the test to check the input is between 0 and 255
if (userInput >=0) and (userInput <256):
    # complete the line to call getHex subprogram
    print(getHex(userInput))
else:
    print("number out of range")
```

Type MS · Subprograms and strings: complete the function and its call

Answer 6 of 6 · Question 2 part c.ii · 4 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c.ii in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark